

- 1) Which of the following is a factor of the expression?

$$x^2 + 6x - 16$$

- A. $(x + 4)$
- B. $(x - 2)$
- C. $(x + 4)$
- D. $(x - 8)$

- 2) Which of the following is a factor of the expression?

$$4x^2 - 4x - 3$$

- A. $(x - 3)$
- B. $(2x + 3)$
- C. $(2x + 1)$
- D. $(2x - 1)$

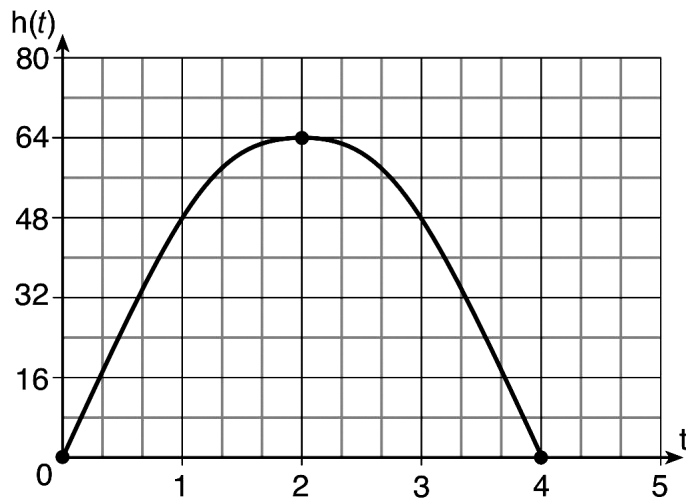
- 3) Which of the following is a factor of the expression?
 $5x^2 + 3x - 8$

- 4) Which of the following is a factor of the expression?
 $3x^2 - 13x + 12$

- 5) Which of the following is a factor of the expression?
 $6x^2 + x - 12$

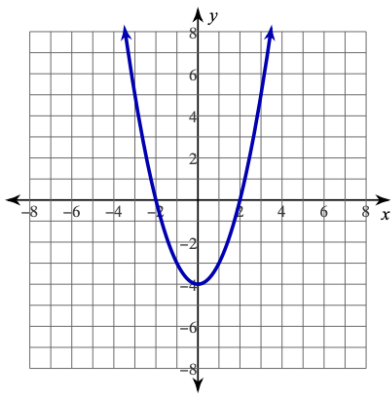
- 6) Which of the following is a factor of the expression?
 $x^2 - 2x - 24$

- 7) The diagram below shows the graph of h , which models the height, in feet, of a rocket t seconds after it was shot into the air.

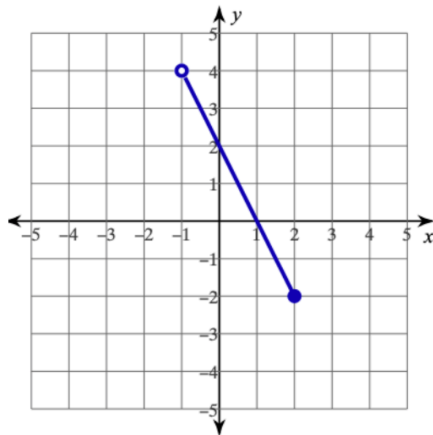


The range of h is

- 8) What is the domain and range of the function graphed below?

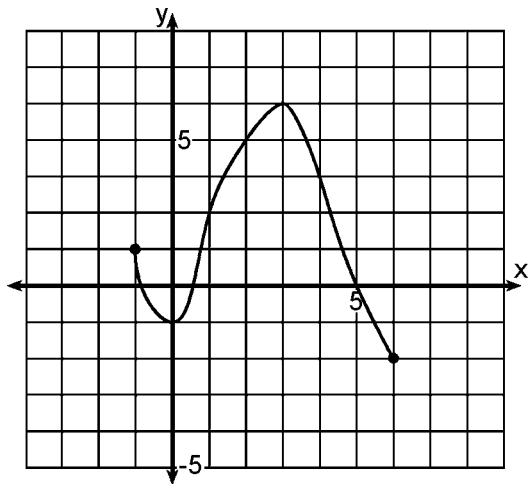


- 9) A function, f , is represented by the line segment drawn in the accompanying graph.

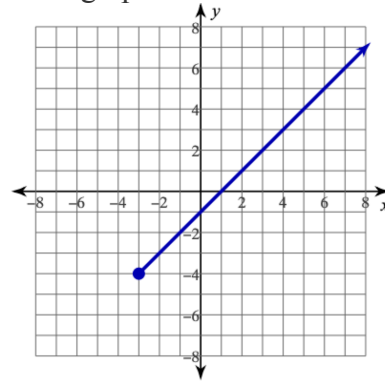


What is the domain and range of function f ?

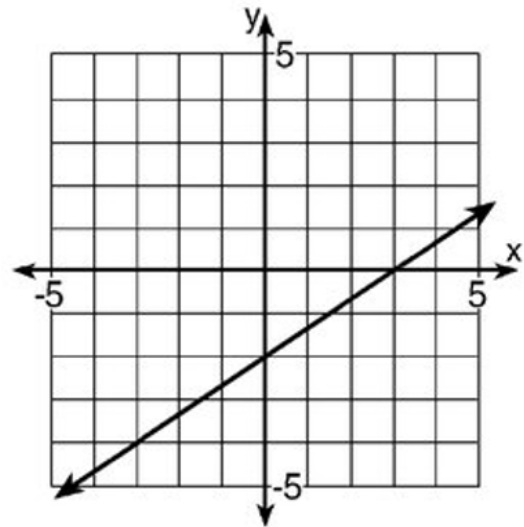
- 10) What is the domain of the function shown below?



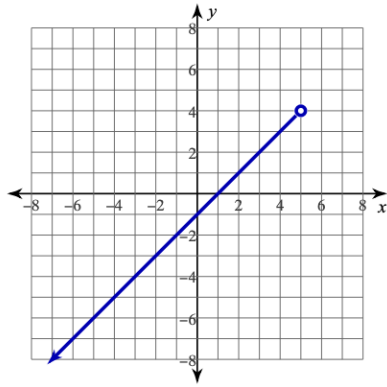
- 11) Find the domain and range of the function shown in the graph.



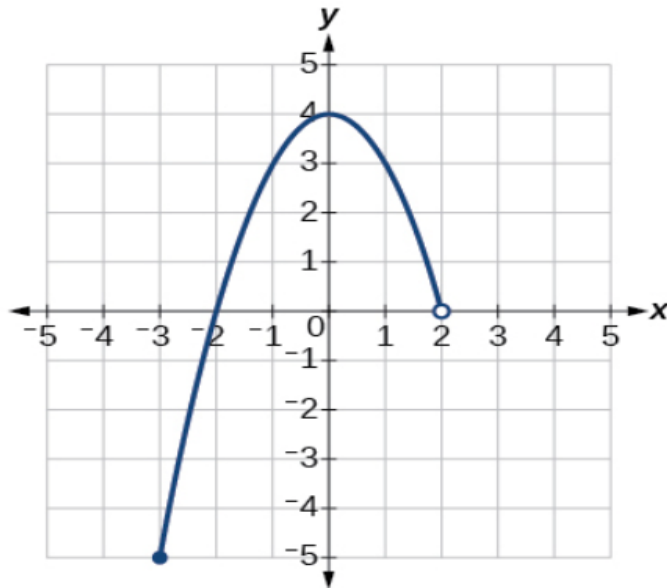
- 12) What is the domain and range of the function graphed below:



13) What is the domain and range of the function graphed below?



14) What the domain and range of the function represented in the graph below?



15) Simplify:

$$\sqrt{150}$$

A) $6\sqrt{2}$

B) $6\sqrt{3}$

C) $5\sqrt{6}$

D) $12\sqrt{2}$

16) Simplify:

$$\sqrt{54x^2y^4}$$

A) $3xy\sqrt{2xy}$

B) $9xy^2\sqrt{6}$

C) $3y^2x\sqrt{6}$

D) $9x^2y^4\sqrt{6}$

17) Simplify

$$\sqrt{27x^3y^4}$$

A) $3y^2x\sqrt{3x}$

B) $9xy\sqrt{3xy}$

C) $3y^2\sqrt{3x}$

D) $9y^2x\sqrt{3}$

18) Evaluate the expression:

$$7 + 3 \cdot 2^2 - 2 - 5$$

19) Simplify the expression:

$$36 \div 3^2 \cdot 5 - 25 + 1$$

20) "Three less than twice z " is represented by the algebraic expression

21) "8 more than the product of 3 and c " is represented by the algebraic expression

22) What is "seven less than five times the sum of a and b " represented as an algebraic expression?

23) What is the value of x in the equation $\frac{3}{4}x + 2 = \frac{5}{4}x - 6$?

24) What is the value of x in the equation
 $5(2x - 7) = 15x - 10$?

25) What is the value of x in the equation
 $13x - 2(x + 4) = 8x + 1$?

26) Solve the equation for x :

$$7(x + 1) + 8 = 2(x + 6) + 5x - 4$$

27) Solve the equation: $5(x + 1) - x = 6x - 2(x - 3) - 1$

28) Solve the equation for x :

$$2(x - 1) + 4x = -x - 2 + 7x$$

29) Solve for x if $7x + 6y = 4z$

30) Solve for L if $P = 2L + 2W$

31) What is the solution to the inequality
 $18 - x < -13$?

32) Solve the inequality $\frac{1}{2}x + 3 < 2x - 6$

Questions 33 and 34 refer to the following:

Solve the given inequality for the domain of the set of real numbers:

33) $5x - 18 > 2(4x - 15)$

34) $-5(3 - 2x) - 59 > 3(x + 1)$

Question 35 refers to the following:

Solve the given inequality for the domain of the set of real numbers:

35) $-5(3 - 2x) - 5 > 10(x + 1)$

36) Solve the inequality: $-4 < 2x - 10 \leq 12$

37) Solve and graph the solution set of the inequality:

$3x + 5 > 5$ or $-2x - 1 \geq 7$

Question 38 refers to the following:

Determine the solution set and graph for the given inequality:

38) $12 < -8x + 4$

39) Solve, and graph the solution set for the inequality $-\frac{3}{4}x + 6 > 3$

40) *Speedy Boat Rental* charges a \$15 deposit fee plus \$2 for each hour of use to rent a paddleboat. What is an equation that can be used to represent C , the total cost to rent the boat, for h hours?

41) The *Roaming Cell Phone Company* charges \$15.00 per month plus \$0.20 per minute of airtime usage. If Juanita uses x minutes of airtime, what is an equation that can be used to determine her monthly cell phone bill (C)?

- 42) A hot air balloon is 1,000 feet above the ground. Randy throws a ball out of the balloon. The ball falls at a constant rate of 32 feet per second. Which of the following equations gives the height (h) of the ball above the ground in terms of seconds (t)?

- 43) Andrea has a sum of money in a savings account. If she withdraws \$150, she still has *at least* \$525 in her savings account. Which inequality can be used to find the minimum amount of money in the account?

- 44) One-third of a number is 12 less than the number itself. If x represents the number, what is an equation that can be used to determine the number?

- 45) The cost of apples is posted in Mr. Riordan's store.

Number of Apples	1	2	3	4
Cost (in cents)	20	40	60	80

This data expresses cost as a function of the number of apples. What is the set that represents the range of this function?

- 46) The cost of oranges is displayed on a poster at a supermarket. The data expresses cost as a function of the number of oranges.

Number of Oranges	1	2	3	4
Cost	0.30	0.60	0.90	1.20

What is the domain of this function?

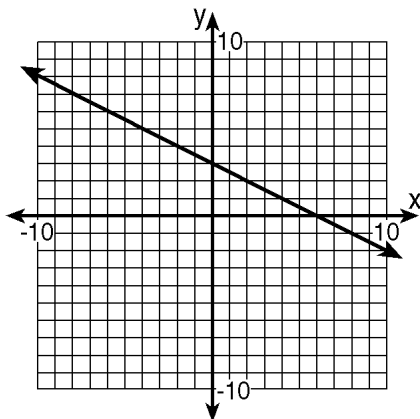
Question 47 refers to the following:

The table below defines the function f .

x	$f(x)$
-1	5
0	1
1	-3
2	-7

47) According to the function shown, what is the value of $f(-1) + f(1)$?

48) The linear function f is defined by the graph below.



What is the value of $f(4)$ in the graph?

Question 49 refers to the following:

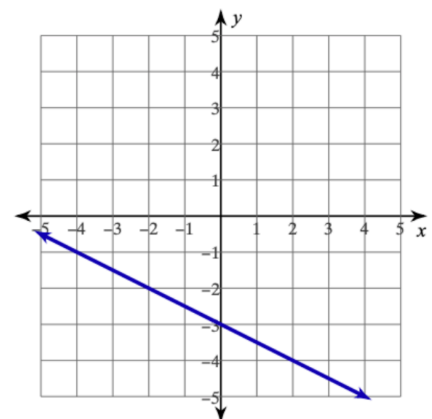
The table below defines the function f .

x	$f(x)$
-1	5
0	1
1	-3
2	-7

49) For what value of x does $f(x) = f(0) - f(2)$ in the function shown?

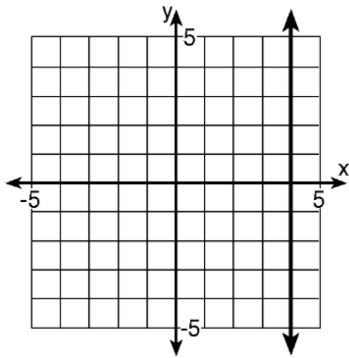
50) If $g(x) = 2x^3 - 3x + 5$, find the value of $g(-2)$.

51) The graph of $f(x)$ is shown below. For what value of x is $f(x) = -4$?



52) If $f(x) = \frac{1}{2}x^2$, what does $f(1) + f(-1)$ equal?

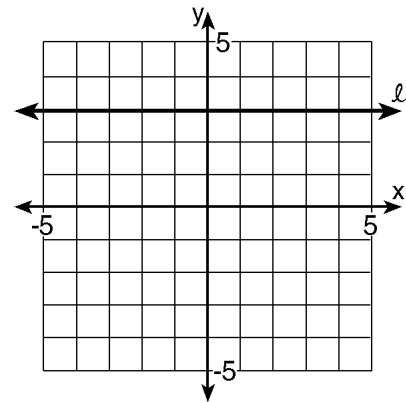
53) What is the slope of the line in this graph?



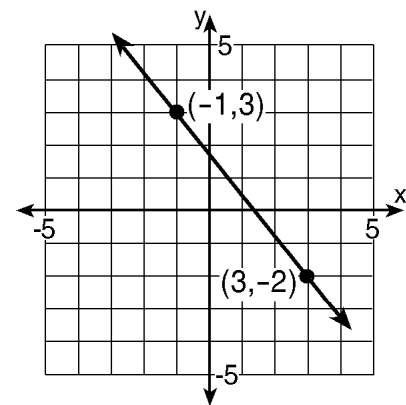
54) What is the slope of the line that passes through the points $(3,2)$ and $(1,7)$?

55) What is the slope of the line that passes through $(2,4)$ and $(-3,4)$?

56) What is the slope of the line in this graph?



57) What is the slope of the line in the accompanying diagram?



58) For the line given by $3y = 2x - 12$, identify the *slope* and *y-intercept*.

59) What is the slope and the y-coordinate of the y-intercept of the equation $2x - 4y = 12$?

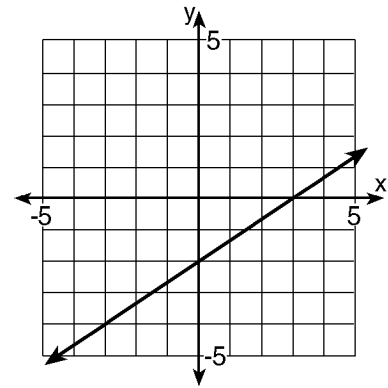
60) What is an equation of a line that is ***perpendicular*** to $y = -2x + 7$ and whose y-intercept is $(0, -4)$?

61) What is an equation of a line that is ***parallel*** to $y = 5x - 1$ and whose y-intercept is $(0, 3)$?

62) What is the y-intercept of the line whose equation is $2x = 3(y - 4) + 3$?

63) What is the x-intercept of the line whose equation is $-17x - y = -85$?

64) What is an equation of the linear function graphed below?



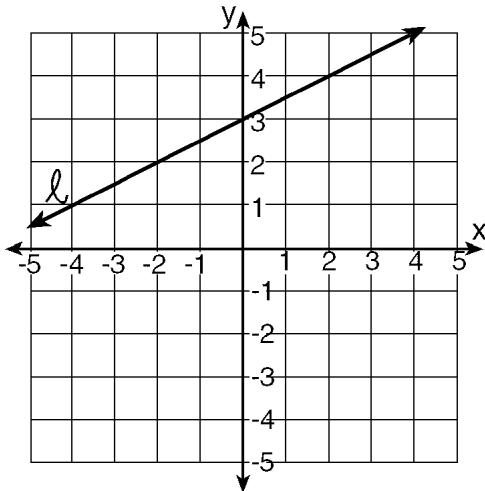
65) What is an equation of a line that has a slope of $-\frac{2}{5}$ and passes through the point $(5, 7)$?

66) A line passes through the points $(7, 4)$ and $(-5, -2)$. What is an equation that *best* defines this line?

67) What is an equation of a line with an x-intercept of $(-2, 0)$ and y-intercept of $(0, 5)$?

68) What is an equation of a line that contains the points $(-2, -2)$ and $(1, 3)$?

69) What equation represents line ℓ , shown in the accompanying diagram?



70) The table below shows the number of hours, x , and the total distance, y , in miles, during a trip.

Time Driving (Hours)	Total Distance (Miles)
0	0
1	55
2	120
3	188
4	252
5	307
6	366

What is an equation that *best* represents the line of best fit for this set of data?

71) Which of the following ordered pairs are solutions to the inequality $3x - 2y \leq 6$?

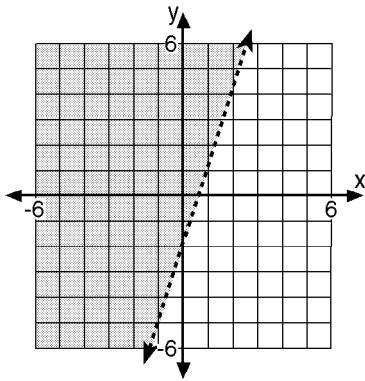
I. $(-3, 3)$

II. $(0, -1)$

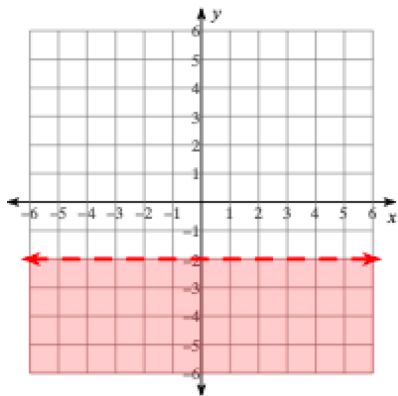
III. $(2, 4)$

IV. $(3, 0)$

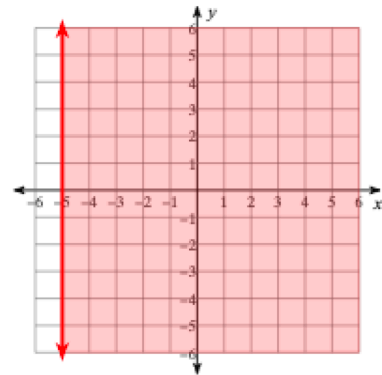
- 72) What inequality is represented by the graph below?



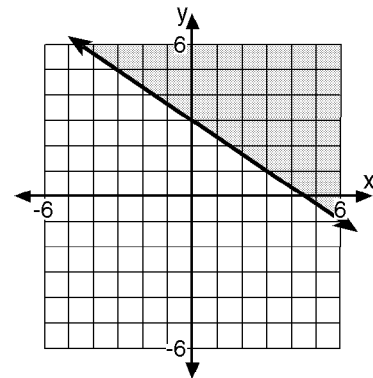
- 73) What inequality is represented by graph below?



- 74) Write an inequality that can be represented by the graph below.



- 75) What inequality is represented by the graph below?



- 76) The total volume of a tree increases 8% each year. What will its volume be after 7 years if its volume is 5 cubic meters now?

Mixed Topics - Multiple Choice Section

- 77) Piper bought herself a new bike using a monthly payment plan. The table below shows the balance after the first 4 payments.

Payments (p)	Balance (b)
1	700
2	600
3	500
4	400

Write an equations that accurately represents the information in this table.

- 78) The accompanying table describes the relationship between the variables x and y .

x	y
1	3,406
3	3,150
5	2,894
7	2,638
9	2,382
11	2,126

Write an equation that can be used to represent this relationship?

- 79) Determine whether $(-1, 4)$ is a solution of the system of equations below.

$$\begin{aligned} -10x + y &= 14 \\ 5x + 4y &= 11 \end{aligned}$$

80) What is the value of y in the following system of equations?

$$2x + 3y = 6$$

$$2x + y = -2$$

81) What is the product of $(-3xy^2)(5x^2y^3)$?

82) The product of $\frac{1}{2}x^3$ and $\frac{1}{4}x^5$ is

83) Simplify $(-4x^3y^2)^4$

84) Simplify $(3x^3y^4)^2$

85) The expression $\sqrt{50}$ can be simplified to

86) Simplify $\frac{-24x^6y^5z^{15}}{4x^3z^5}$?

87) Simplify the expression $\frac{(6x^3)(2x^4)}{2x^3}$?

88) Simplify the expression $\sqrt{75}$

89) Simplify the expression $\sqrt{48} + \sqrt{27}$

90) Simplify the expression $\sqrt{50} + \sqrt{32}$

Question 91 refers to the following:

Simplify the given expression [*Show all work.*]:

91) $\sqrt{27} + \sqrt{12} - \sqrt{48}$

Questions 92 and 93 refer to the following:

Multiply the given expression:

92) $\sqrt{6} \cdot \sqrt{3}$

93) $\sqrt{4} \cdot \sqrt{9}$

94) Simplify the expression $7\sqrt{2} - \sqrt{18}$

Question 95 refers to the following:

Simplify the given expression:

95) $\frac{\sqrt{48}}{\sqrt{3}}$

Question 96 refers to the following:

Express the given difference as a polynomial in standard form:

96) $(5x^2 - 9x + 3) - (-4x^2 + 3x - 5)$

97) Simplify the expression $(2x^2 - 3x + 4) - (-4x^2 + 5x - 6)$

- 98) The accompanying table shows the cost of menu choices at a local food truck. Each price is represented as a polynomial expression.

Menu Choices	Cost (\$)
Hamburger	$3x^2 + 5x - 9$
Hotdog	$x^2 - 3x - 7$
Soda	$4x^2 - 1$
French fries	$x^2 + 6x + 11$

Miranda purchased one order of fries, one soda, and one other item from the menu. What was the third item she selected from the menu if her total purchase cost $8x^2 + 11x + 1$? [*Show work to support your answer.*]

- 99) What is the product of $(x - 8)(x + 9)$?

- 100) Simplify the expression $(x - 5)^2$

- 101) What is the greatest common factor of $12x^2 - 8x + 6$

Question 102 refers to the following:

Represent the given product of the two binomial expressions as the sum of monomials:

102) $(5y - 2)(y - 4)$

Questions 103 and 104 refer to the following:

Use the distributive law write the given algebraic expression as a polynomial in standard form [*Show all work.*]:

103) $(x - 3)(x^2 + x + 3)$

104) $(2x^2 + 3x - 4)(2x - 5)$

105) Find the product:
 $(2x^2 - 3x + 4)(x^2 + 5x - 2)$

106) What is the greatest common factor of $20y^2 - 4y$?

107) Factor $x^2 - 81$?

108) Factor $4x^2 - 25$?

109) Factor $x^3 - 5x^2 - 9x + 45$?

110) If $3x$ is one factor of $3x^2 - 9x$, what is the other factor?

111) Factor:
 $y^2 + 4y - 32$

112) Factor $x^3 + 7x^2 - 36x + 253$?

113) Factor:
 $y^2 + 8y + 15$

114) What are the coordinates of the *minimum* point of the quadratic function whose equation is $f(x) = x^2 + 3$?

115) What is the value of x in the equation $3x^2 = 7x$?

116) What is the value of x in the equation $x^2 + 5x = 6$?

117) What are the solutions to the equation $y^2 + y - 12 = 0$?

118) What are the solutions to the equation $x^2 + 3x - 6 = 12$?

119) The solution set for the equation $x^2 - 2x - 15 = 0$ is

120) $x^2 - 25 = 0$

121) $3x^2 = 27$

122) $4(x + 3)^2 = 64$

123) $2(x + 3)^2 - 64 = 0$

124) The function $f(x) = 5x^2 + 20x + 14$ will have a

Questions 120 through 123 refer to the following:

Solve the given quadratic equation by taking a square root and express irrational roots in radical form [*Show all work.*]:

125) The quadratic function $f(x) = -x^2 + 4x + 1$ will have a

126) What is an equation of the axis of symmetry of the graph of the function $f(x) = 2x^2 - 3x - 1$?

127) What is an equation of the axis of symmetry of the quadratic function whose equation is $f(x) = 3x^2 - 12x - 13$?

128) What are the coordinates of the vertex of the parabola whose equation is $y = 2x^2 + 4x - 3$?

129) What are the coordinates of the vertex of the quadratic function whose equation is $f(x) = x^2 - 4x + 4$?

130) What is the y-intercept of the parabola whose equation is $y = x^2 + 5x - 6$?

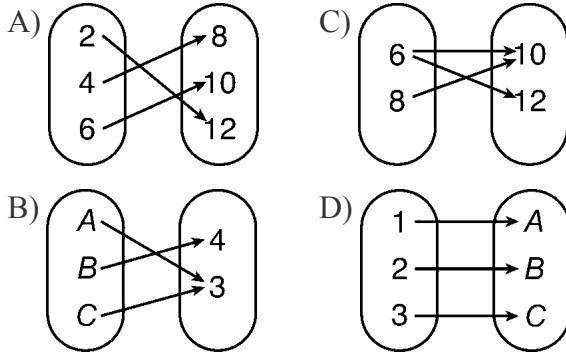
131) Write an equation that models the data in the table below.

x	0	1	2	3
y	2	8	32	128

132) The approximate population growth of a certain bacteria is represented by the function $f(t) = 5(3)^t$. What is the population when $t = 4$?

133) Rhonda's car is expected to depreciate at a rate of 20% per year. If her car is currently valued at \$28,000, to the nearest dollar, how much will it be worth in five years?

134) Which relation is *not* a function?

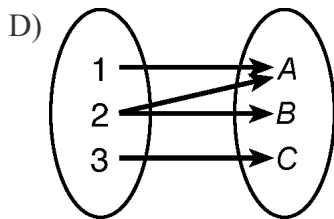
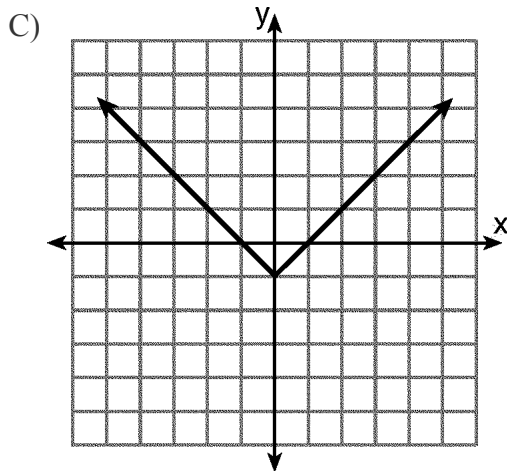


135) Which one of the following relations is *not* a function?

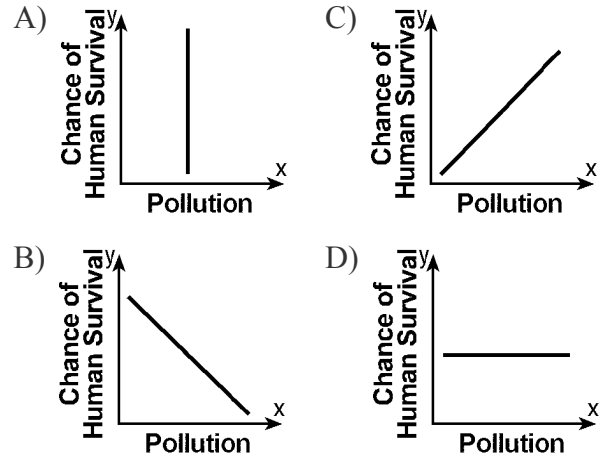
A) $3x + 2y = 4$

B)

x	y
-10	-2
-6	2
-2	6
1	9
5	13



136) Which graph does *not* represent a function of x ?



137) Which one of the following tables represents a function?

A)

x	3	5	7	9
$f(x)$	2	4	2	4

B)

x	0	-1	0	1
$f(x)$	0	1	-1	0

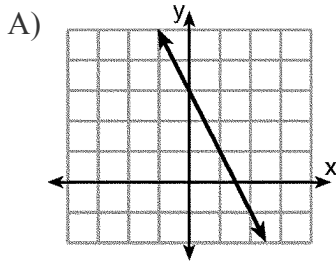
C)

x	2	4	2	4
$f(x)$	3	5	7	9

D)

x	0	1	-1	0
$f(x)$	0	-1	0	1

- 138) Which one of the following functions has a constant rate of change equal to -3 ?



B) $\{(1,5), (2,2), (3,-5), (4,4)\}$

C) $2y = -6x + 10$

D)

x	y
0	2
1	5
2	8
3	11

- 139) Which one of the following lines contains the point $(-2, -4)$?

A) $y = -\frac{1}{2}x - 4$

C) $3x - 2y = 5$

B) $y = 3x + 2$

D) $x - 2y = -10$

- 140) Which one of the following charts could represent the function $f(x) = -2x + 6$?

A)

x	f(x)
0	4
2	6
4	8
6	10

C)

x	f(x)
0	8
2	10
4	12
6	14

B)

x	f(x)
0	6
2	10
4	14
6	18

D)

x	f(x)
0	6
2	2
4	-2
6	-6

- 141) Which one of the following charts represents solutions for the equation $y = \frac{1}{2}x + 3$?

A)

x	2	3	4	5
y	-2	0	2	4

B)

x	-2	0	2	4
y	1	3	5	7

C)

x	-2	0	2	4
y	2	3	4	5

D)

x	-2	0	2	4
y	0.5	1.5	2.5	3.5

- 142) Which table of values represents a linear relationship?

A)

x	f(x)
-1	-1
0	0
1	1
2	8
3	27

C)

x	f(x)
-1	-3
0	-2
1	1
2	6
3	13

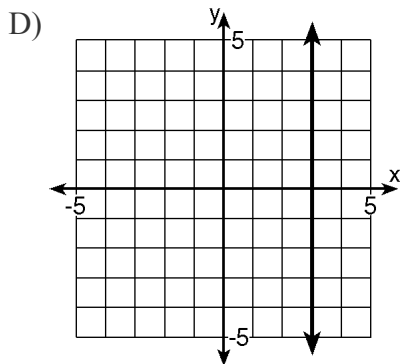
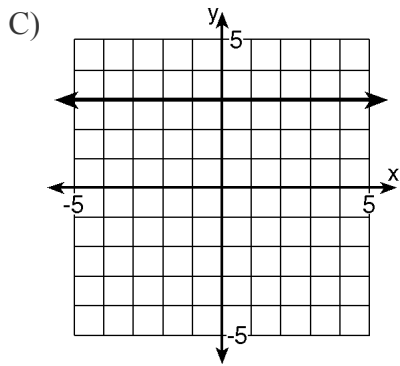
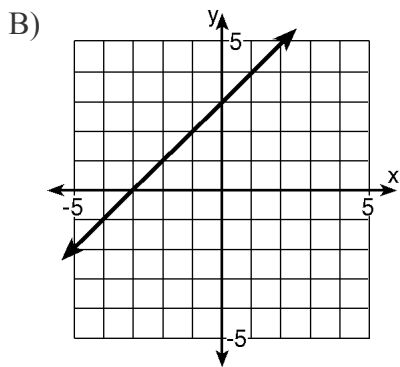
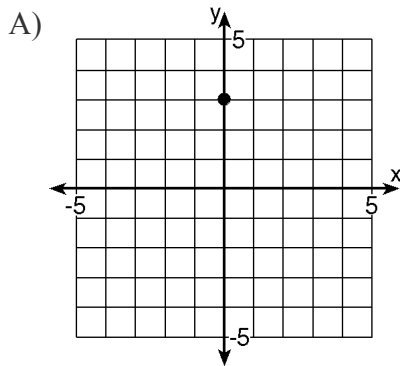
B)

x	f(x)
-1	$\frac{1}{2}$
0	1
1	2
2	4
3	8

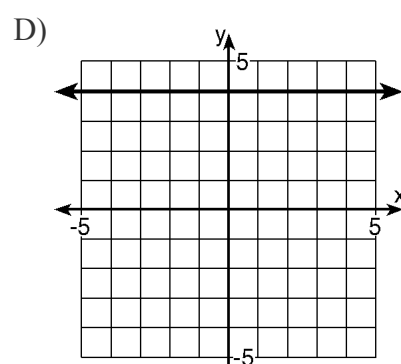
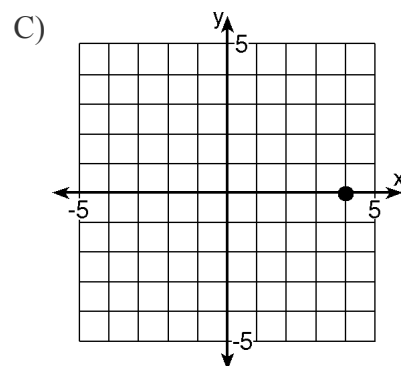
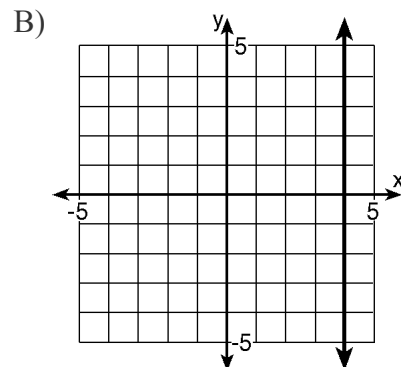
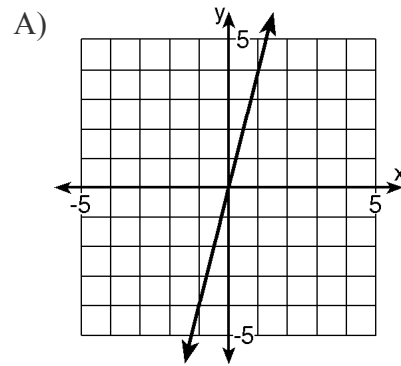
D)

x	f(x)
-1	-3
0	-1
1	1
2	3
3	5

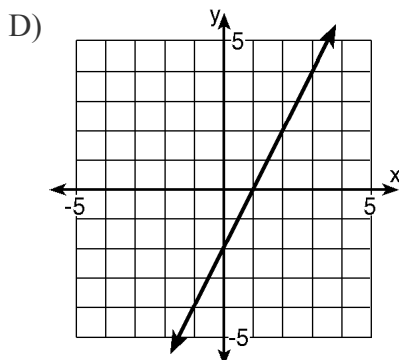
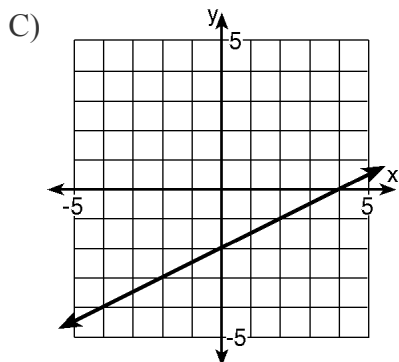
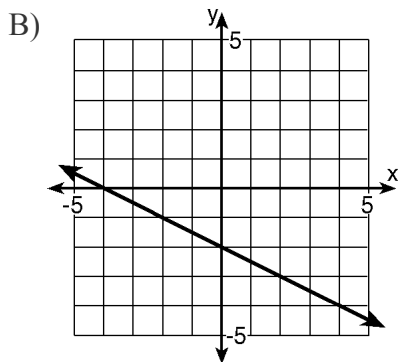
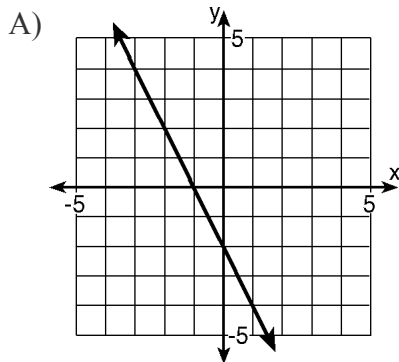
143) Which one of the following graphs *best* represents the equation $f(x) = 3$?



144) Which one of the following graphs is the *best* representation of the function $x = 4$?



- 145) Which one of the following graphs *best* represents the equation $y = -\frac{1}{2}x - 2$?



- 146) Rowan has \$50 in a savings jar and is putting in \$5 every week. Jonah has \$10 in his own jar and is putting in \$15 every week. Each of them plots his progress on a graph with time on the horizontal axis and amount in the jar on the vertical axis. Which one of the following statements about their graphs is true?

- A) Rowan's graph has a steeper slope than Jonah's.
- B) Rowan's graph always lies above Jonah's.
- C) Jonah's graph always lies above Rowan's.
- D) Jonah's graph has a steeper slope than Rowan's.

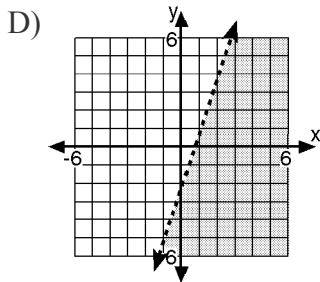
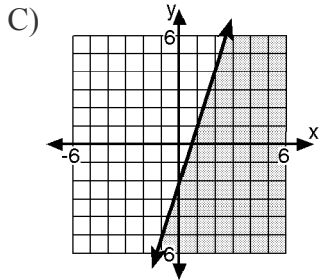
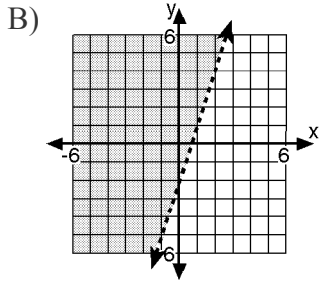
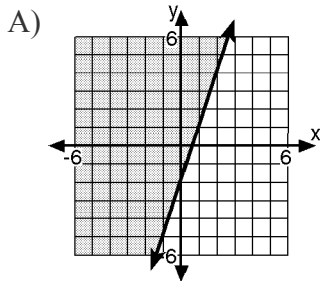
- 147) The cost of airing a commercial on television is modeled by the function $C(n) = 110n + 900$, where n is the number of times the commercial is aired. Based on this model, which statement is true?

- A) The commercial costs \$900 to produce and \$110 each time it is aired.
- B) The commercial costs \$0 to produce and \$110 per airing up to \$900.
- C) The commercial costs \$1,010 to produce and can air an unlimited number of times.
- D) The commercial costs \$110 to produce and \$900 each time it is aired.

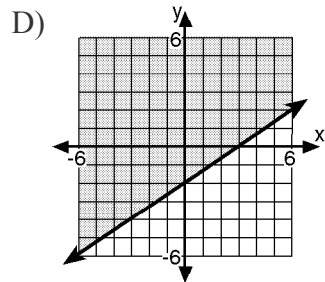
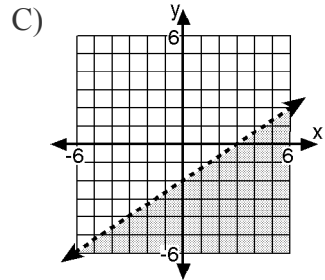
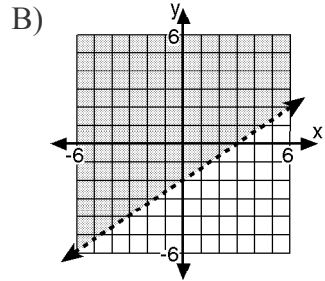
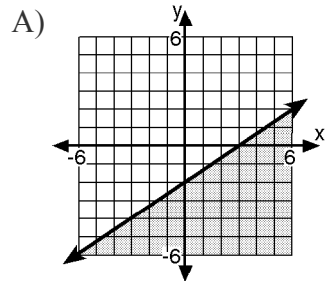
- 148) Which ordered pair is *not* in the solution set of $y > 2x + 1$?

- A) (2,5) B) (1,4) C) (3,8) D) (1,6)

149) Which of the following is the graph of $y \leq 3x - 2$?



150) Which of the following is the graph of $2x - 3y > 6$?



151) The perimeter of a rectangular garden is 42 feet. The length of the garden is 5 feet more than twice the width. Which system of equations will determine the length (ℓ) and the width (w) of the garden?

A) $2\ell + w = 42$
 $2w + \ell = 5$

C) $2\ell + w = 42$
 $2\ell - w = 5$

B) $2\ell + 2w = 42$
 $\ell = 2w + 5$

D) $2\ell + 2w = 42$
 $w = 2\ell - 5$

- 152) The length of a rectangle is 5 feet more than the width. The perimeter is 22 feet. Which system of equations below will determine the length (ℓ) in feet and the width (w) in feet of the rectangle?

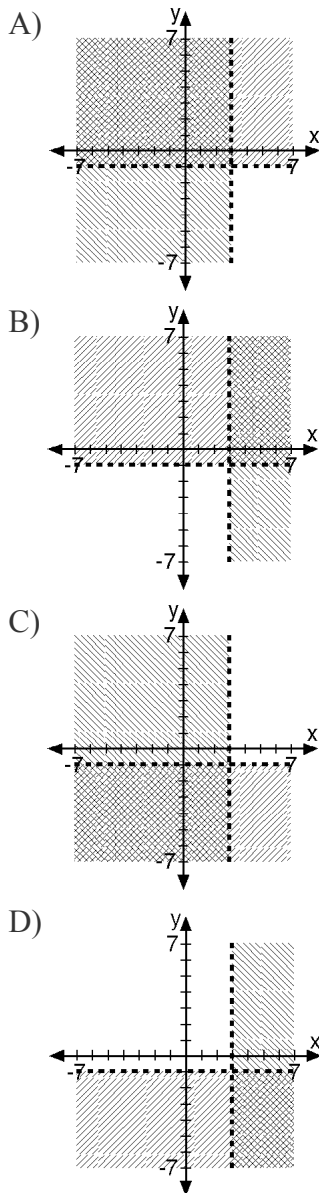
A) $\ell - w = 5$
 $\ell + w = 22$

C) $\ell = 5w$
 $22 = \ell + w$

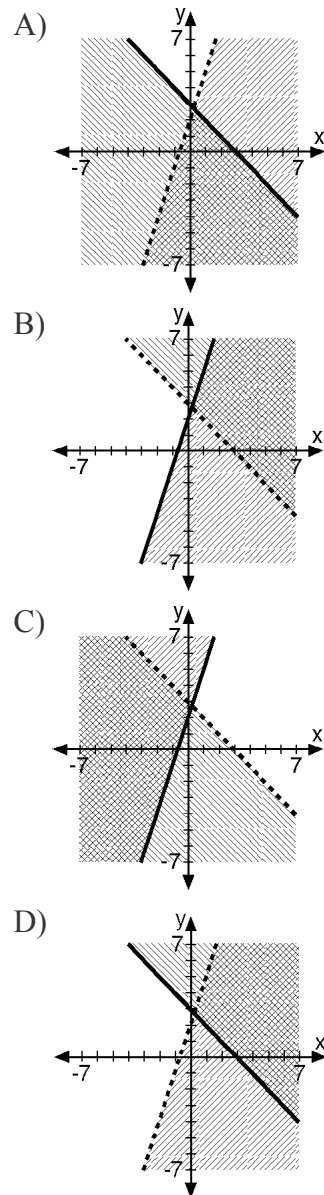
B) $\ell = w + 5$
 $2\ell + 2w = 22$

D) $w = \ell + 5$
 $2\ell + 2w = 22$

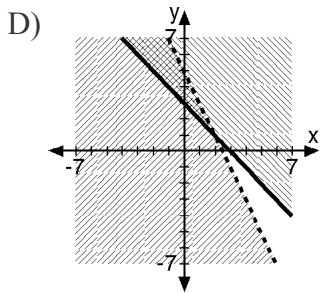
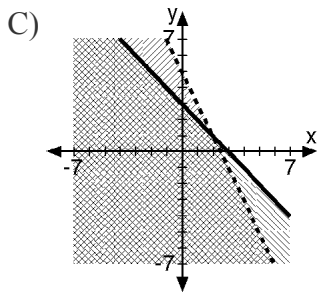
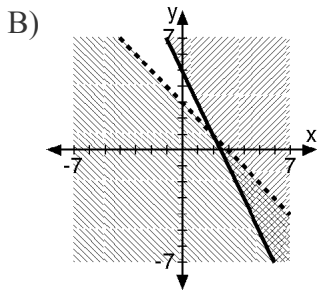
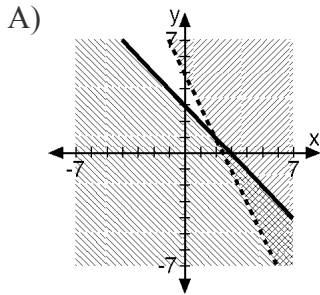
- 153) Which of the following graphs represents the solution to the system of equations, $x > 3$ and $y < -1$?



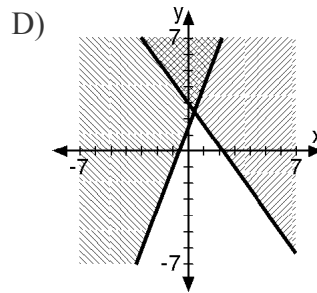
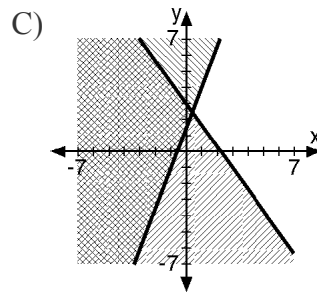
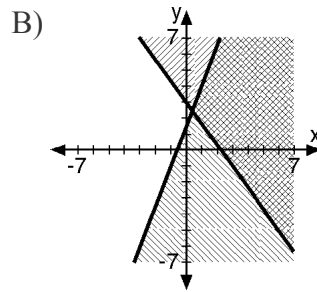
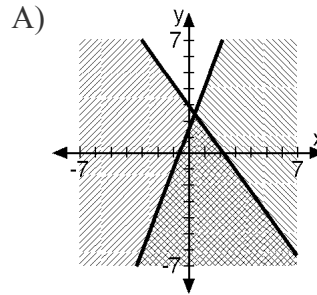
- 154) Which of the following graphs represents the solution to the system of equations, $y < 3x + 2$ and $y \geq -x + 3$?



- 155) Which of the following graphs represents the solution to the system of equations, $2x + y > 5$ and $x + y - 3 \leq 0$?



- 156) Which of the following graphs represents the solution to the system of equations, $5x - 2y + 3 \geq 0$ and $4x + 3y - 12 \leq 0$?



- 157) The point $(-3, 5)$ is located on the graph of which of the following equations?

- | | |
|------------------|-----------------|
| A) $x + y = 2$ | C) $y = x - 1$ |
| B) $2x + 3y = 6$ | D) $x + y = -2$ |

- 158) Which one of the following is an example of linear growth?
- A) A bacterial culture doubles in size each day
 - B) A baby gains 0.5% of its weight each week
 - C) An investment of \$100 at an annual interest rate of 0.75%.
 - D) Depositing \$5.00 in a savings account each week for t weeks

- 159) Which one of the following is an example of exponential growth?
- A) Investing \$1,000 at an annual interest rate of 3%.
 - B) A bacterial culture increases by 1,000 bacteria per day
 - C) The height of a tree increases by 2 feet each year
 - D) Water drips into a bucket at a rate of 2 pints each minute

- 160) Which equation models the data in the accompanying table?

Time in Hours (x)	0	1	2	3	4	5	6
Population (y)	5	10	20	40	80	160	320

- A) $y = 2x$
- B) $y = 2^x$
- C) $y = 5(2^x)$
- D) $y = 2x + 5$